

C++ Programming Concepts in Inheritance

Data Abstraction

Hiding Complex Implementation Details & Showing Only Essential Information



Access Modifiers in Inheritance

public

Accessible by Derived Class.

private

Not Inherited by Derived Class.

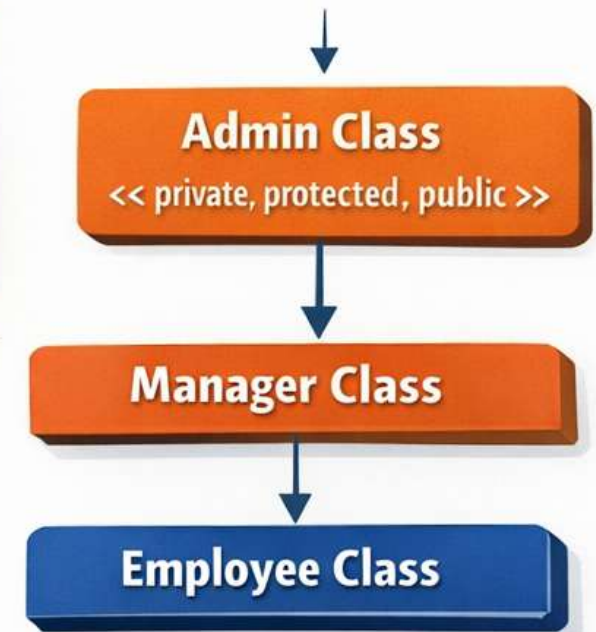
protected

Accessible within Derived Class.

Visibility Modifiers in Inheritance

Base Class	public	private	protected
public	public	private	protected
private	Not Inherited	Not Inherited	Not Inherited
protected	protected	private	protected

Example Class Hierarchy



q1.cpp > Admin

```
1  #include<iostream>
2  using namespace std;
3
4  class Admin{
5      private:
6          bool can_terminate;
7          double total_annual_revenue;
8          string company_name;
9
10         protected:
11             double manager_salary;
12             double employee_salary;
13             int total_staff;
14
15         public:
16             Admin(){
17                 company_name = "Tata Consultancy Services(TCS)";
18                 total_annual_revenue = 900000;
19                 total_staff = 150;
20                 can_terminate = true;
21                 manager_salary = 70000;
22                 employee_salary = 50000;
23             }
24
25             virtual void myAccess(){
26                 cout<<"\n.....Admin Access.....\n";
27                 cout<<"Company Name: "<<company_name<<endl;
28                 cout<<"Manager Salary: "<<manager_salary<<endl;
29                 cout<<"Employee Salary: "<<employee_salary<<endl;
30                 cout<<"total staff: "<<total_staff<<endl;
31                 cout<<"total anual revenue: "<<total_annual_revenue<<endl;
32                 cout<<"can terminate: "<<can_terminate<<endl;
33             }
34         }
35
36
37     };
```

```
G+ q1.cpp > main()
4   class Admin{
38
39   class Manager:public Admin{
40       public:
41           void myAccess()override{
42               cout<<"\n.....Manager Access.....\n";
43               cout<<"Manager Salary: "<<manager_salary<<endl;
44               cout<<"Employee Salary: "<<employee_salary<<endl;
45               cout<<"total staff: "<<total_staff<<endl;
46           }
47       };
48   };
49
50   class Employee:public Manager{
51       public:
52           void myAccess()override{
53               cout<<"\n.....Employee Access.....\n";
54               cout<<"Employee Salary: "<<employee_salary<<endl;
55               cout<<"Total staff: "<<total_staff<<endl;
56           }
57   };
58   int main(){
59       Admin a;
60       Manager m;
61       Employee emp;
62
63       a.myAccess();
64       m.myAccess();
65       emp.myAccess();
66
67       return 0;
68   }
```

.....Admin Access.....

Company Name: Tata Consultancy Services(TCS)

Manager Salary: 70000

Employee Salary: 50000

total staff: 150

total anual revenue: 900000

can terminate: 1

.....Manager Access.....

Manager Salary: 70000

Employee Salary: 50000

total staff: 150

.....Employee Access.....

Employee Salary: 50000

Total staff: 150

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